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Hominin habitats

The section [Background and beginnings in the Miocene](#) describes certain global climatic changes that reduced forested areas and induced more open terrestrial [biomes](#) during the late [Miocene Epoch](#) (11.2–5.3 mya). During the succeeding [Pliocene Epoch](#) (5.3–2.6 mya) these changes only intensified. In [Africa](#), [primates](#) diversified. In [Eurasia](#), contrarily, [hominins](#) disappeared by the beginning of the Pliocene. The only descendants of Late Miocene primates in [Asia](#) are the extinct Early-Middle [Pleistocene](#) *Gigantopithecus blacki* of southern [China](#) and northern Vietnam and the present-day [orangutans](#) and gibbons of South and [Southeast Asia](#).

MAJOR SITES OF AUSTRALOPITH FOSSIL FINDS IN AFRICA

▲ *Australopithecus*

● *Kenyanthropus*

■ *Orrorin*

● *Paranthropus*

✕ *Ardipithecus*



hominid fossil sites

A selection of locations in sub-Saharan Africa where hominid fossils have been found.

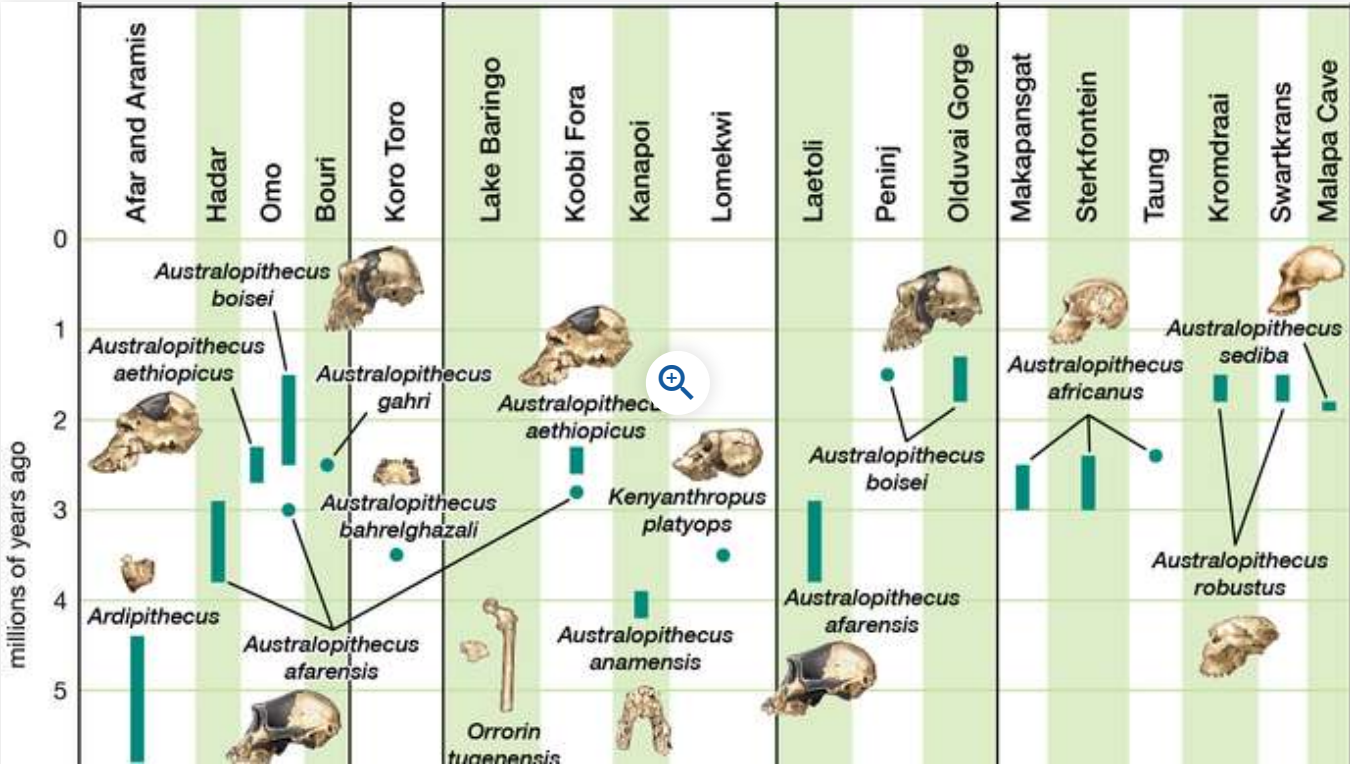
Image: Encyclopædia Britannica, Inc.

It is reasonable to expect that the increased variety and shifting distribution of African biomes stimulated new [hominin](#) lifeways, some of which led to survival and others of which did not. Insofar as [habitats](#) have been (or can be) discerned from evidence found with the Pliocene hominin species, hominins inhabited a variety of biomes in eastern, central, and southern Africa. In central [Ethiopia](#), *Ar. ramidus* is associated with faunal and floral remains indicating a woodland habitat. Later remains, in northern Ethiopia, indicate *Au. afarensis* inhabited a mosaic of riverine [forest](#), lowland woodland, [savanna](#), and dry bushland. In northern [Kenya](#) *Au. anamensis* lived in dry open woodland or bushland with a gallery forest along a nearby [river](#). In central [Chad](#) the northernmost and westernmost species, *Au. bahrelghazali*, appears to have lived in a mosaic of open and wooded biomes near a river. Mammalian [fossils](#) from Lomekwi, northern Kenya, indicate that *Kenyanthropus platyops* inhabited a relatively well-watered area of forest or closed woodland or the forest edge between them. The habitat of the 3.5-million-year-old [Laetoli](#) hominins in northern [Tanzania](#) was arguably a mosaic of open [grassland](#) and more-closed woodland. The area may have been wetter than it is now. No permanent water source has been identified for the Laetoli area during the Pliocene. Later in the Pliocene, *Au. garhi* was active on broad, grassy plains bordering a lake in central Ethiopia. [Models](#) of the habitat of *Au. africanus*, based on fauna from the two major South African [cave](#) sites—[Sterkfontein](#) and [Makapansgat](#)—stress closed-canopy wooded conditions: either dry woodland with [grasslands](#) nearby or subtropical forest. During the [tenures](#) of *H. habilis* and *P. boisei* at [Olduvai Gorge](#), northern Tanzania, the climate changed from moist to dry and again to moist before a long dry span that began two million years ago. Specimens of both of these Olduvai hominins are mostly from the shore of an ancient saline, alkaline [lake](#). At [Koobi Fora](#), northern Kenya, specimens of *H. habilis* have been more commonly found in lake-margin deposits, while those of *P. boisei* are equally common in river and lake-margin sediments. Fossil [pollen](#) indicates that highland forest was nearby and that near the lake there were grassy areas and dense woodland and [shrubland](#).



At Konso, southern Ethiopia, *P. boisei* lived in a grassland habitat. Elsewhere in [eastern Africa](#), *P. aethiopicus* was associated with closed habitats. The South African

cave sites (Swartkrans, Kromdraai, and Drimolen) of *P. robustus* are associated with open and even arid habitats, but these may not reflect its actual foraging preference.



australopith fossil locations

Approximate time ranges of sites yielding australopith fossils.

Image: Encyclopædia Britannica, Inc.

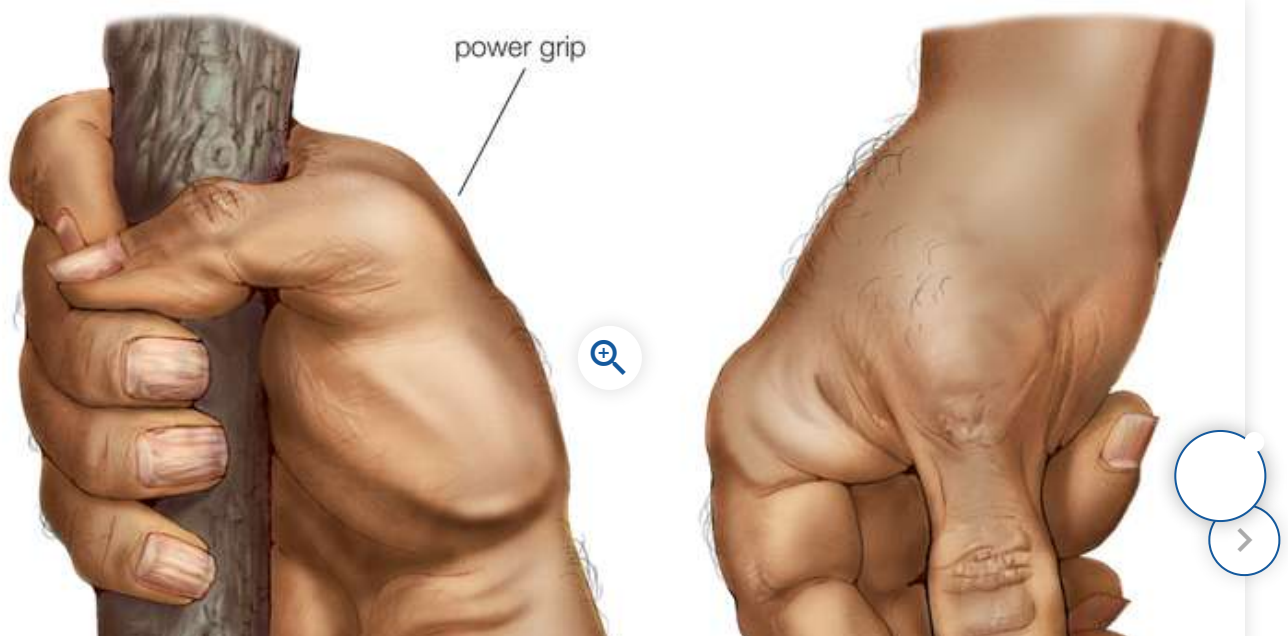


One of the more profound effects of **Pliocene** habitat changes was honing the energy-conservant bipedal stride at the time that *Homo* species deployed out of Africa and into Eurasia. Shortly after *Homo* evolved in Africa, some species ventured to temperate biomes in Eurasia and then to subtropical and tropical biomes in South and Southeast Asia. Subsequently there was a **migration** back to Africa, perhaps as early as 1.8–0.9 mya. This hemispheric dispersion of *Homo* is associated with elaboration of stone **tool** kits, increased **brain** size, and reduction in size of the **jaws** and **teeth**—all of which are the subject of the next section.

Tools, hands, and heads in the Pliocene and Pleistocene

Refinements in **hand structure**

Primates are hand-to-mouth feeders that pluck and catch items selectively by hand before ingesting them. Without **tools**, emergent **hominins** would have relied on the versatility and strength of their hands to collect food and on their **teeth** and **jaws** alone to process it. Unless they used tools to fashion carrying devices such as bags from **animal skins**, they would have needed a reliable source of **water** nearby, and they would also have been limited in the types and number of objects that they could transport through their range. In addition to transporting objects and water, there is the more obvious utility of animal skins in protecting against night chills, **rain**, and strong sunshine.





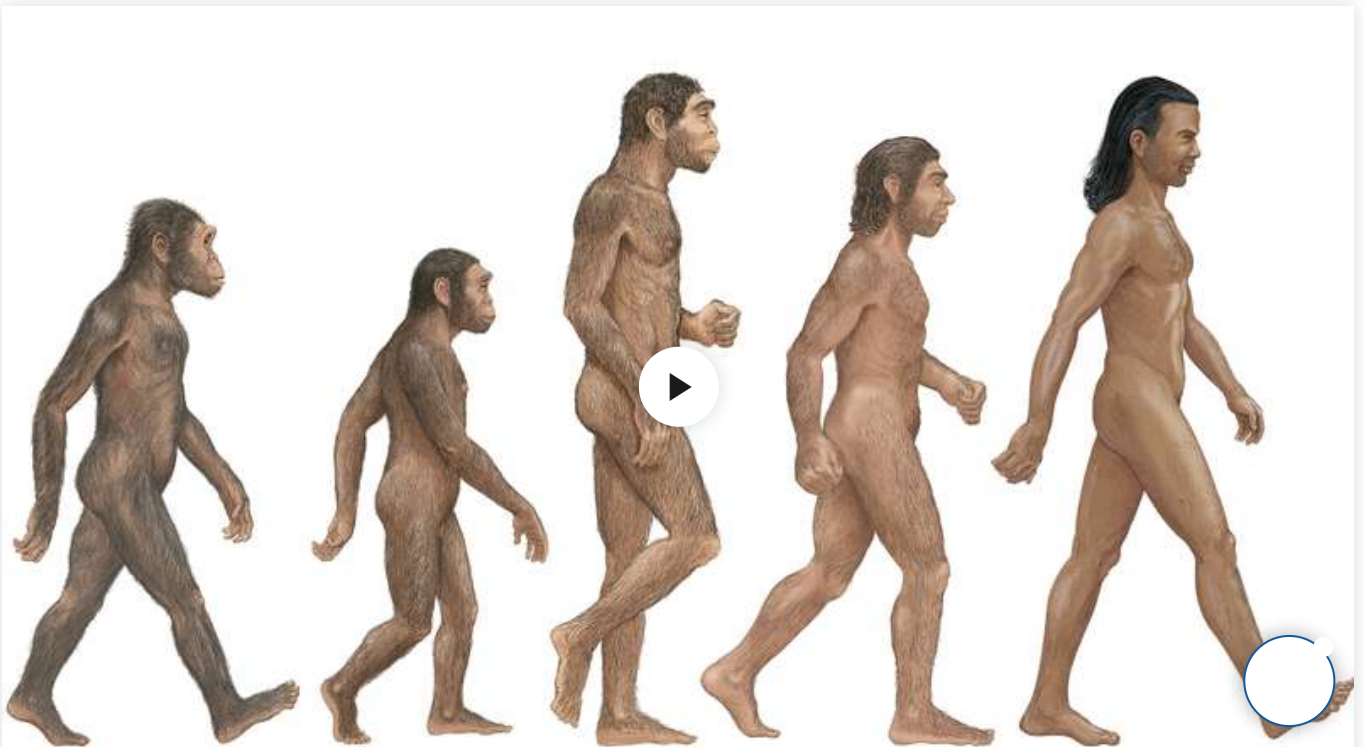
power and precision grips

A fully opposable thumb gives the human hand its unique power grip (left) and precision grip (right).

Image: Encyclopædia Britannica, Inc.

Sharp-edged **stones**, even small flakes, would be a boon to early hominins who learned how to select and make them for cutting hides, meat, sticks, and other plant material. Stones also would assist in pounding open hard-shelled **fruits** and **nuts**, **bones** for **marrow**, and **skulls** for brains. There may have been a span when early hominins used naturally occurring stones and other objects as tools and weapons, much as some wild **chimpanzees** do today.

Before hominins controlled fire and either built sturdy shelters on the ground or effectively defended caves and rock shelters, they may have constructed platforms in trees for daily activities as well as night lodging. Raw materials, stone **hammers**, cutting tools, and sticks and stones for defense could be stored in the trees to be used repeatedly. Handheld rocks, **clubs**, and long stabbing sticks, **spears**, or other missiles would constitute a formidable defense, especially if employed from the vantage of a tree platform.



Compare Homo habilis, H. erectus, H. neanderthalensis, and H. sapiens to determine

the first human species

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HUMAN EVOLUTION

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By about 3.3 million years ago, some hominins were making and using simple stone artifacts in eastern [Africa](#). The tools—primitive hammers, [anvils](#), and cutting tools—predate the [emergence](#) of the oldest confirmed specimens of *Homo* by nearly 1 million years, and paleontologists speculate that they were likely constructed by members of *Australopithecus* or *Kenyanthropus*, who then inhabited the region.

Because the earliest stone artifacts were of such simple construction and because chimpanzees, [orangutans](#), and [capuchin monkeys](#) today can employ stones, stems, vines, and sticks to extract nutritious morsels from protective covers, one need not expect that early hominin toolmakers displayed modern hand structure and exquisite motor control. Nonetheless, the unique structure of the [human](#) hand is readily explained by a substantial history of producing and using increasingly complex tool kits and other artifacts. (Attributing specific advancements in artifact manufacture to the threefold increase in brain size between [Pliocene](#) hominins and *H. sapiens* is a much more difficult hypothesis to support, as will be discussed later in this section.)

The features of human [hands](#) are easily distinguishable from those of the great [apes](#), and they underpin our refined manipulatory abilities. The most complex [adaptations](#) of the human hand involve the [thumb](#), wherein a unique, fully independent [muscle](#) (the flexor pollicis longus) gives this digit remarkable strength in pinch and power [grips](#). The fingertips are broad and equipped with highly sensitive pads of [skin](#). The proportional lengths of the thumb and other [fingers](#) give us an opposable thumb with precise, firm contact between its tip and the ends of each of the other fingers. A special saddle joint and associated ligaments at the base of the thumb facilitate refined rotation. Special configurations of joints at the bases of the fifth, fourth, and second fingers facilitate tip-to-tip precision grips with the thumb. Asymmetry of the heads of the second and fifth palm bones induces rotation of the articulated fingers during opposition with the thumb. Finally, numerous modifications of the small

muscles in the hand are associated with fine control of the thumb and fingers.

Au. afarensis is the earliest hominin species for which there are sufficient fossil hand bones to assess manipulatory capabilities. They were capable of gripping sticks and stones firmly for vigorous pounding and throwing, but they lacked a fully developed human power grip that would allow cylindrical objects to be held between the partly flexed fingers and the palm, with counterpressure being applied by the thumb. There are insufficient specimens to assess fine manipulation in *Australopithecus*, but there is no reason to believe that they were less capable than modern chimpanzees. Chimpanzees and other apes have remarkable precision of grip, even though the tapered thumb tip must be pressed against the side of the index finger and cannot be apposed securely to any of the fingertips.

Hand bones assigned to a 1.8-million-year-old specimen of *H. habilis* from Olduvai Gorge in northern Tanzania represent an advance over those of *A. afarensis* in features related to tool use. Tools similar to those found at Olduvai are found associated with *H. habilis* from other parts of eastern Africa as well. The tips of its thumb and fingers were flat, and there is evidence for a strong flexor pollicis longus muscle and a saddle joint at the base of the thumb. Hand bones arguably assigned to *P. robustus* or *Homo* from Swartkrans, South Africa, confirm that by about 1.8 mya one or more hominin species had highly developed thumbs and flat fingertips.

Hominin hand bones from 2.8–2.5-million-year-old cave deposits at Sterkfontein, South Africa, may be evidence that the hands of *A. africanus* were somewhat more advanced for stone tool use, but no artifact has been found in association with them. Younger Sterkfontein deposits (2.0–1.5 mya) contain stone artifacts and remains of a *Homo* species.

Because of an absence of fossils, it is not possible to track certain refinements in hand structure that must have evolved in conjunction with innovations in tool manufacture and use during the heydays of *H. rudolfensis*, *H. ergaster* (1.9–1.5 mya), and *H. erectus* (1.7–0.2 mya), as well as *H. antecessor* (1.0–0.8 mya) and *H. heidelbergensis* (600–200 kya). Only prehistoric and modern *H. sapiens* and *H. neanderthalensis* are fully represented by hand skeletons.

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